

ANALYSTLAB AFRICA CONSULTING

DATASET INSPECTION REPORT

**ABC Communications Ltd
Customer Churn & Retention Analysis**

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1. Purpose of the Inspection

The purpose of the dataset inspection was to establish whether the customer churn dataset was sufficiently structured and prepared for business analysis. The inspection focused on the dataset structure, variables, data types, missing or blank values, duplicate records, and numerical fields required for the subsequent analysis.

2. Dataset Overview

The dataset used for the analysis is the Telco Customer Churn dataset supplied for the case study. It contains customer-level information covering customer identification, demographics, tenure, services, contract arrangements, payment methods, monthly charges, total charges and churn status.

Inspection item	Result
Number of customer records	7,043
Unit of analysis	Individual customer
Outcome variable	Churn (Yes / No)
Customer identifier	customerID
Key numerical fields	tenure, MonthlyCharges, TotalCharges
Key categorical fields	Contract, InternetService, PaymentMethod and other customer/service attributes
Analytical churn field	Churn_Numeric: 1 = churned, 0 = stayed

3. Rows, Columns and Data Structure

The data were inspected at customer-record level. Each row represents a customer and the columns describe the customer's characteristics, subscribed services, contractual arrangement, payment method, charges and churn outcome.

The analysis retained the original dataset while creating cleaned analytical fields where necessary. This approach preserves the source data and provides separate fields suitable for numerical calculations and correlation analysis.

4. Data Types Inspection

Field / field group	Expected analytical type	Inspection / treatment
customerID	Text	Retained as customer identifier.
tenure	Numeric / integer	Used for tenure analysis and correlation.
MonthlyCharges	Numeric / decimal	Cleaned analytical version used where the source formatting was not consistently recognised as numeric.
TotalCharges	Numeric / decimal	Blank-looking/text-formatted values were investigated and a cleaned analytical version was used.
Churn	Categorical text	Original Yes/No values retained.
Churn_Numeric	Binary numeric	Created for correlation: 1 = churned, 0 = stayed.
Contract	Categorical text	Used for contract-level churn analysis.
InternetService	Categorical text	Used for service-level churn analysis.
PaymentMethod	Categorical text	Used for payment-method churn analysis.

5. Missing and Blank Value Inspection

Blank-looking values were specifically investigated in the charge fields, particularly TotalCharges. These values can cause Excel to treat a numerical field as text and can result in #VALUE! errors when mathematical functions are applied.

During preparation, cleaned numerical fields were created so that valid numerical observations could be used consistently in averages, distributions and correlation calculations.

The final report does not state an exact missing-value count because that exact count was not established in the completed inspection work. The inspection therefore records the issue and the treatment applied without inventing a figure.

6. Duplicate Record Inspection

Customer IDs were checked as part of the data-quality inspection because customerID is intended to identify individual customer records. Duplicate checking was included before the business analysis to reduce the risk of counting the same customer more than once.

No specific duplicate count is stated here because the completed analysis did not establish a final verified duplicate figure. The inspection process should be retained in the Excel workbook as supporting evidence.

7. Data Cleaning and Preparation

Original source fields were retained to preserve the raw data.

MonthlyCharges was prepared in a cleaned numerical field for calculations where the original formatting was not consistently interpreted as numeric.

TotalCharges was prepared in a cleaned numerical field to address text/blank formatting issues.

Churn_Numeric was created to convert the categorical churn outcome into a numerical binary variable, with 1 representing churn and 0 representing customers who stayed.

Text values were inspected for inconsistent spacing or formatting before PivotTable analysis.

The cleaned fields were then used for statistical analysis, including averages and the correlation heatmap.

8. Key Numerical Summary Findings

Measure	Result	Business interpretation
Average MonthlyCharges — stayed	R60.81	Customers who stayed had a lower average monthly charge.
Average MonthlyCharges — churned	R76.28	Churned customers had a higher average monthly charge.

Tenure ↔ Churn correlation	-0.3522	Moderate negative association: longer tenure is associated with lower observed churn.
MonthlyCharges ↔ Churn correlation	0.1934	Weak positive association.
TotalCharges ↔ Churn correlation	-0.1983	Weak negative association.

9. Data Readiness Assessment

Following inspection and preparation, the dataset was considered suitable for the required business analysis. The cleaned numerical fields allowed the analysis to proceed without relying on inconsistent text-formatted charge values, while Churn_Numeric enabled numerical correlation analysis.

The inspection also highlighted an important data-quality consideration: Excel can interpret apparently numerical values as text when formatting or blanks are present. Cleaning and validating numerical fields before analysis is therefore essential to avoid calculation errors and misleading results.

10. Link to Business Analysis

The inspected dataset was subsequently used to produce the business analysis required by the assignment, including bar charts, pie charts, histograms, a box plot and a correlation heatmap. The inspection findings provide the data-quality foundation for interpreting those outputs.

11. Conclusion

The dataset inspection confirmed that the customer churn data contain the necessary customer, service, contract, payment and financial variables for churn analysis. The principal preparation requirement was ensuring that charge fields were treated as numerical data and that churn was represented numerically for correlation analysis. The cleaned dataset was therefore suitable for the subsequent business analytics and customer-retention recommendations.